AN EXPLICIT FERMIONIC LAUGHLIN GAP ON THE SPHERE FOR EVERY PARTICLE NUMBER
A REMARK ON OPENAI'S "A FOCK-SPACE INEQUALITY AND THE LAUGHLIN SPECTRAL GAP"

[AUTHOR NAME]

Abstract. OpenAI's preprint [1] proves $H_Q^2 \geq \gamma H_Q$ on the fermionic Fock space of the spherical V_1 Hamiltonian for every $\gamma < \gamma_* = 0.046167$ and all $Q \geq Q_\gamma$, with Q_γ non-explicit, and deduces the Laughlin gap $1/25$ for $N \geq N_0$. We replace its only non-effective step (a transfer lemma) by a positivity check at the given flux, and certify every flux in ball arithmetic. Result: $H_Q^2 \geq 0.05947\, H_Q$ for every integer $Q \geq 24$, and $H_{N,3(N-1)} \geq 0.05947\,(I - P_{\mathrm{L}})$ for every $N \geq 2$. All ideas except Lemma 5 and the bookkeeping of Proposition 3 are OpenAI's.

1. Statement

We use the notation of [1]: U_Q is spin $Q/2$ with lowering basis $e_0, \dots, e_Q$; $V_Q \subset \bigwedge^2 U_Q$ is the spin-$(Q-1)$ summand with lowering basis v_p, $v_0 = -e_0 \wedge e_1$; $B_p = B(v_p)$ and $H_Q = \sum_{p=0}^{2Q-2} B_p^\dagger B_p$ on $\mathcal{F}_Q = \bigoplus_N \bigwedge^N U_Q$ (coefficient one per unordered pair projector; this is `LaughlinFock.hamiltonian`).

Theorem 1. *For every integer $Q \geq 24$, $H_Q^2 \geq 0.05947\, H_Q$ on $\mathcal{F}_Q$. More precisely $H_Q^2 \geq \gamma_Q H_Q$ with $\gamma_Q \geq 0.136071$ (24 $\leq Q \leq$ 48), ≥ 0.166963 (49 $\leq Q \leq$ 72), ≥ 0.079313 (73 $\leq Q \leq$ 299), ≥ 0.059470 ($Q \geq$ 300).*

Theorem 2. *For every $N \geq 2$ and $Q = 3(N-1)$, $H_{N,Q} \geq 0.05947\,(I - P_{\mathrm{L},N})$ on $\bigwedge^N U_Q$.*

Theorem 1 gives the Comparator statement `thm_fock` with the explicit threshold $Q_\gamma = 24$ for every $0 < \gamma < \gamma_*$ (since $H^2 - \gamma H = (H^2 - \gamma_Q H) + (\gamma_Q - \gamma)H$). Theorem 2 is [1, Cor. 1.2] with $N_0 = 2$ and constant $0.05947 > 1/25$.

Proof of Theorem 2 from Theorem 1. For $N \geq 9$, $Q = 3(N-1) \geq 24$, so $H_{N,Q}^2 \geq \gamma H_{N,Q}$ and $\operatorname{spec} H_{N,Q} \subset \{0\} \cup [\gamma, \infty)$. By [1, Lemma 6.2], valid for all $N \geq 2$, the kernel is $\mathbb{C}\Psi_{\mathrm{L},N}$. For $2 \leq N \leq 8$ see §4. □

2. The comparison at a fixed flux

Fix $Q \geq 24$ and the support $\mathcal{S}_t = \{(p,j) : p \leq 7,\ j \leq 8,\ 0 \leq p + j - t \leq 8\}$, $t = 0, \dots, 7$, as in [1, eq. (F)]. Let $\Gamma_t = F_t F_t^{\mathsf{T}}$ with F_t rational, indexed by $\{B_t\} \cup \mathcal{S}_t$, and $K = \sum_t \sum_{m,m'} (\Gamma_t)_{mm'} O_m^\dagger O_{m'}$ with $O_{B_t} = B_t$, $O_{(p,j)} = c_{p+j-t}^\dagger c_j B_p$. Thus K is a sum of squares, and $\mathcal{K}_Q = (2Q-1) \int K(g)\, dg \geq 0$. The paper's seven rows are the case $\operatorname{rank} \Gamma_t \leq 1$. Let $\eta = \sum_t (\Gamma_t)_{00}$, $d_a = 2Q-1$, $d_{3,z} = 3Q-1-2z$, $d_D = 4Q-1-2D$, and let $q_z(Q)$, e_z^Q, E_D^Q, S_r^Q, χ, $n_D = \lfloor (D+1)^2/4 \rfloor$ be as in [1, §§3–5]. All of these are evaluated at the given Q, with no limits.

Proposition 3. *Let $s_z = q_z(Q) - (d_a/d_{3,z}) e_z^Q$ for $z \in \{2, 4, 5, \dots, 15\}$, $c_3 = \sum_z \max(0, -s_z)(z+1) d_{3,z}/d_a$, $\delta_Q = \sum_{17 \leq z \leq Q,\ z \text{ odd}} (d_{3,z}/d_a) |q_z(Q)| (z+1)$, and $C_D = (d_D/d_a)\chi - E_D^Q$. Suppose that*

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for each $1 \le D \le 23$ *there is* $\rho_D \ge 0$ *with*

$$\widehat{W}_b\big((S^Q)^{\mathsf T} C_D S^Q + \rho_D I\big)\widehat{W}_b^{\dagger} \;\ge\; 0 \quad \text{on } \textstyle\bigwedge^4 U_Q, \tag{1}$$

where S^Q *is the* $\mathcal{R}_D \times \{b=(p,j,k) : p+j+k=D,\ j<k\}$ *matrix of the highest vectors* $|D,D;r\rangle_Q$ *in the orthonormal vectors* $v_p \otimes (e_j \wedge e_k)$*, and* $\widehat{W}_b$ *maps* $b \mapsto v_p \wedge e_j \wedge e_k$*. Then* $H_Q^2 - \gamma H_Q \ge \mathcal{K}_Q \ge 0$ *on* $\mathcal{F}_Q$ *with* $\gamma = 1-\eta-\delta_Q-c_3-\sum_D n_D\rho_D$*.*

Proof. By [1, Lemma 2.1], $H^2 = H + \mathcal{L}_3(W_3(W_3^\dagger W_3 - I)W_3^\dagger) + \mathcal{L}_4(W_4 W_4^\dagger)$. By the normal ordering [1, (normal2)–(normal4)] and Schur averaging [1, §4], $\mathcal{K}_Q = \eta H + A_3 + A_4$, where A_3 is given by [1, (A3)] and A_4 by [1, (A4)]. Their derivations hold at every $Q \ge 24$, because all indices stay inside the representations.

Three body. Blocks $z = 0, 1, 3$ are killed by W_3 [1, Lemma 3.2]. In block $z \le 15$ the difference of coefficients is s_z. If $s_z < 0$, the argument of [1, Lemma 6.1] (Cauchy–Schwarz on the highest vector and averaging) gives $\mathcal{L}_3(W_3 R_{3,z} W_3^\dagger) \le (z+1)(d_{3,z}/d_a)H$. Blocks $z \ge 16$ have $e_z^Q = 0$. Even ones are positive and odd ones cost at most $\delta_Q H$, by the same lemma.

Four body. In spin block D the target minus A_4 equals $d_a \int \mathcal{L}_4\big(\sum_{r,s}(C_D)_{rs}|w_{Dr}\rangle\langle w_{Ds}|\big)(g)\,dg$ [1, proof of Prop. 5.3], where $w_{Dr} = \widehat{W}_b S^{Q\,\mathsf T} e_r$. Inequality (1) says $\sum_{rs}(C_D)_{rs}|w_{Dr}\rangle\langle w_{Ds}| \ge -\rho_D \sum_b |u_b\rangle\langle u_b|$, $u_b = v_p \wedge e_j \wedge e_k$. Lift by $\mathcal{L}_4$, which preserves positivity, and average. Since $\|B(u_b)\psi\| = \|c_k c_j B_p \psi\| \le \|B_p\psi\|$ and $d_a \int B_p(g)^\dagger B_p(g)\,dg = H$, the right side costs at most $n_D\rho_D H$, exactly as in [1, Prop. 5.3]. Blocks $D \notin [1,23]$ of the target are positive and have no A_4 part. □

Remark 4. This replaces [1, (3certificate), (4certificate), Lemma 5.2, Prop. 5.3] and the "sufficiently large Q" in the proof of [1, Thm. 1.1]. No limiting object and no transfer is used. A crude effectivisation of the transfer error of [1, Lemma 5.2] gives a constant $\ge 10^{65}$ at $Q = 24$ (estimate of the primary session, not re-derived here).

Reductions of (1)**.** Write $\widehat{W}_b = \Delta' N \Lambda''$ with N an integer, Q-independent matrix and Δ', Λ'' positive diagonal. This follows from [1, eq. (pair)], since $v_p(x,y) \propto (x-y)\sqrt{\binom{Q}{x}\binom{Q}{y}}$. Hence $\operatorname{ran}\widehat{W}_b^{\mathsf T} = \Lambda''\operatorname{ran} N^{\mathsf T}$ has an exact rational basis. By the compression lemma [1, Lemma 5.1], (1) is equivalent to (b-space) $B^{\mathsf T}(S^{\mathsf T} C S + \rho I)B > 0$ for a basis B of $\operatorname{ran}\widehat{W}_b^{\mathsf T}$. Minimising $\|y\|$ over $y \in \operatorname{ran}\Pi$ with $S\Pi y = c$ shows that this is also equivalent to (r-space) $\rho G + GCG \ge 0$ with $G = S\Pi S^{\mathsf T}$, Π the projection onto $\operatorname{ran}\widehat{W}_b^{\mathsf T}$. If $|J| = \operatorname{rank} G$ and $G_{JJ} > 0$, the r-space form reduces to the $|J| \times |J|$ matrix $\rho G_{JJ} + G_{J\cdot} C G_{\cdot J} > 0$. This needs $\operatorname{rank} G = \operatorname{rank} W_D$ (here W_D has columns w_{Dr}).

Lemma 5 (Lemma R)**.** *For* $Q \ge 24$ *and* $1 \le D \le 23$*,* $\operatorname{rank} W_D = \operatorname{hw}(D) - \operatorname{zm}(D)$*. Here* $\operatorname{hw}(D) = c(D+1) - c(D)$*, where* $c(n)$ *is the number of* 4*-subsets of* $\mathbb{N}$ *with sum* n*, and* $\operatorname{zm}(D) = p_4(D-17) - p_4(D-18)$*, where* p_4 *counts partitions into at most* 4 *parts.*

Proof. $\widehat{W}$ is an intertwiner and the $|D,D;r\rangle_Q$ span the spin-J highest-weight space of $V_Q \otimes \bigwedge^2 U_Q$, $J = 2Q-1-D$. So $\operatorname{span}\{w_{Dr}\} = \mathrm{HW}_J \cap \operatorname{ran}\widehat{W}$. Now $\psi \perp \operatorname{ran}\widehat{W}$ iff $B(v)\psi = 0$ for all $v \in V_Q$ iff $H_{4,Q}\psi = 0$, so $\operatorname{ran}\widehat{W} = (\ker H_{4,Q})^\perp$ and $\operatorname{rank} W_D = \dim \mathrm{HW}_J(\bigwedge^4) - \dim \mathrm{HW}_J(\ker H_{4,Q})$. Weights of $\bigwedge^4 U_Q$ are index sums. Spin J corresponds to sum $D+1 \le 24 \le Q$, giving $\operatorname{hw}(D)$. The divisibility argument of [1, Lemma 6.2] gives $\ker H_{4,Q} = \prod_{i<j} b_{ij}^3 \cdot \operatorname{Sym}^4 U_{Q-9}$. The prefactor is invariant, so spin J corresponds to deficit $D-17$ in $\operatorname{Sym}^4 U_{Q-9}$, which gives $\operatorname{zm}(D)$. □

The values for $D = 1, \dots, 23$ are $0,0,0,0,1,0,1,1,2,1,3,2,4,3,5,4,6,5,7,6,8,7,9$. For the boxes only the upper bound rank $\le$ hw $-$ zm is needed: a lower rank would be detected by failure of $G_{JJ} > 0$. On copies $r \ge 17$ the matrix C_D vanishes identically ($j+k \le 16$ on the support), and $w_{D,1} = 0$ for even D [1, §4]. So these copies may be dropped.

3. Certificates

Data. Four rational Γ (44-bit dyadic F_t), obtained by rounding float SDP solutions on flux windows $\{24, 36, 48\}$, $\{48, 60, 72\}$, $\{72, 100, 150, 300\}$, $\{300, 1000, 10^8\}$.

Integer $24 \le Q \le 299$. At $x = 1/Q$ exactly, Arb balls (256 bits): η exact; e_z^Q, $q_z(Q)$ and the exact tail; E_D^Q and S^Q from closed-form coupling coefficients in x. Each ρ_D is a rational chosen from a float eigenvalue and then certified by a strict ball LDL^{T} of the r-space matrix and of G_{JJ}. Independently, all 276 values were re-certified in the rank-free b-space form, which does not use Lemma 5.

$Q \ge 300$. Every quantity is analytic in $x \in [0, 1/24]$. We use 168 boxes with rational endpoints forming an exact contiguous cover of $[0, 1/300]$. Each quantity is expanded as $f(x_0 + t) = \sum_{i<6} f_i(x_0)t^i + R$, with the coefficients from Arb power series at the exact centre and $|R| \le |F_6|h^6$, where F_6 is the order-6 series coefficient computed at the ball base point (it encloses $f^{(6)}(\xi)/6!$ on the box). Matrices are bounded below by $T_0 + T_1 t - (\sum_{i\ge 2} \|T_i\|_F h^i)I$, which is affine in t, so it suffices to certify the endpoints $t = \pm h$. The box $[0, 1/50400]$ covers all $Q \ge 50400$, so no limit is needed. The tail uses $z \le 250$ exactly and $\sum_{z>250} 6z(z+1)(2/3)^z$ [1, Lemma 6.1].

4. Small particle numbers

For $2 \le N \le 8$ ($Q \le 21$) we work in the $L_z = 0$ sector, which meets every multiplet because $QN/2 \in \mathbb{Z}$. There H is $\sqrt{\beta(S)\beta(S')}$ times an exact rational matrix. We certify $X = H - gI + \psi\psi^{\mathsf{T}} > 0$ for a float vector ψ by float Cholesky with Higham's backward error bound [3, Thm. 10.3] (with a $10\times$ safety factor). By Courant–Fischer this gives $\lambda_2 \ge g$, and $\lambda_1 = 0$ is simple by [1, Lemma 6.2]. We get $g_N \ge 0.9990, 0.7847, 0.6796, 0.6083, 0.5847, 0.5477, 0.5305$ for $N = 2, \dots, 8$. As an independent recheck we rebuilt H from Clebsch–Gordan coefficients and verified $\#\{\lambda < g_N\} = 1$ by LDL^{T} inertia. Lanczos with residuals gives the same gaps (e.g. $N = 8$: 0.5315767947, residual $4 \cdot 10^{-15}$).

5. Independent checks and limitations

(i) From Clebsch–Gordan coefficients we checked, at $Q = 24, 27$: the pair vectors against [1, eq. (pair)]; the coefficient-one normalisation against first quantisation; [1, Lemma 2.1] on $N = 3, 4$; and $q_z(Q)$. All agree to 10^{-15}. (ii) We tested the full operator inequality $H^2 - \gamma_Q H - \mathcal{K}_Q \ge 0$ directly on every spin sector, using the certified γ_Q and the rational Γ: $(Q, N) \in \{24\} \times \{3, 4, 5\}$, $\{27, 29\} \times \{3, 4\}$, and $Q = 49, 66, 73, 100$ with $N = 3$. Minimum eigenvalue $\ge -2 \cdot 10^{-15}$. A phase-flipped negative control fails by -778. (iii) Lemma 5 was checked by exact rational rank at $Q = 24, 300, 10^3, 10^4, 10^8$ and by float SVD. (iv) For five boxes we re-derived the Taylor coefficients by 13-node interpolation and checked the order-6 enclosures. We also re-certified points and integer fluxes in each box with the box's ρ_D, including the rank-free form.

Not claimed: Lean formalisation; optimality (sector tests suggest $\gamma \approx 0.21$ near $Q = 24$); the planar statement. Related: the bosonic V_0 adaptation [2] certifies $Q \ge 3088$ ($N \ge 1545$), with an analogous stable-rank formula. It leaves smaller Q and N open.

References

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